DQN and Advanced Policy Gradient

Map

Map

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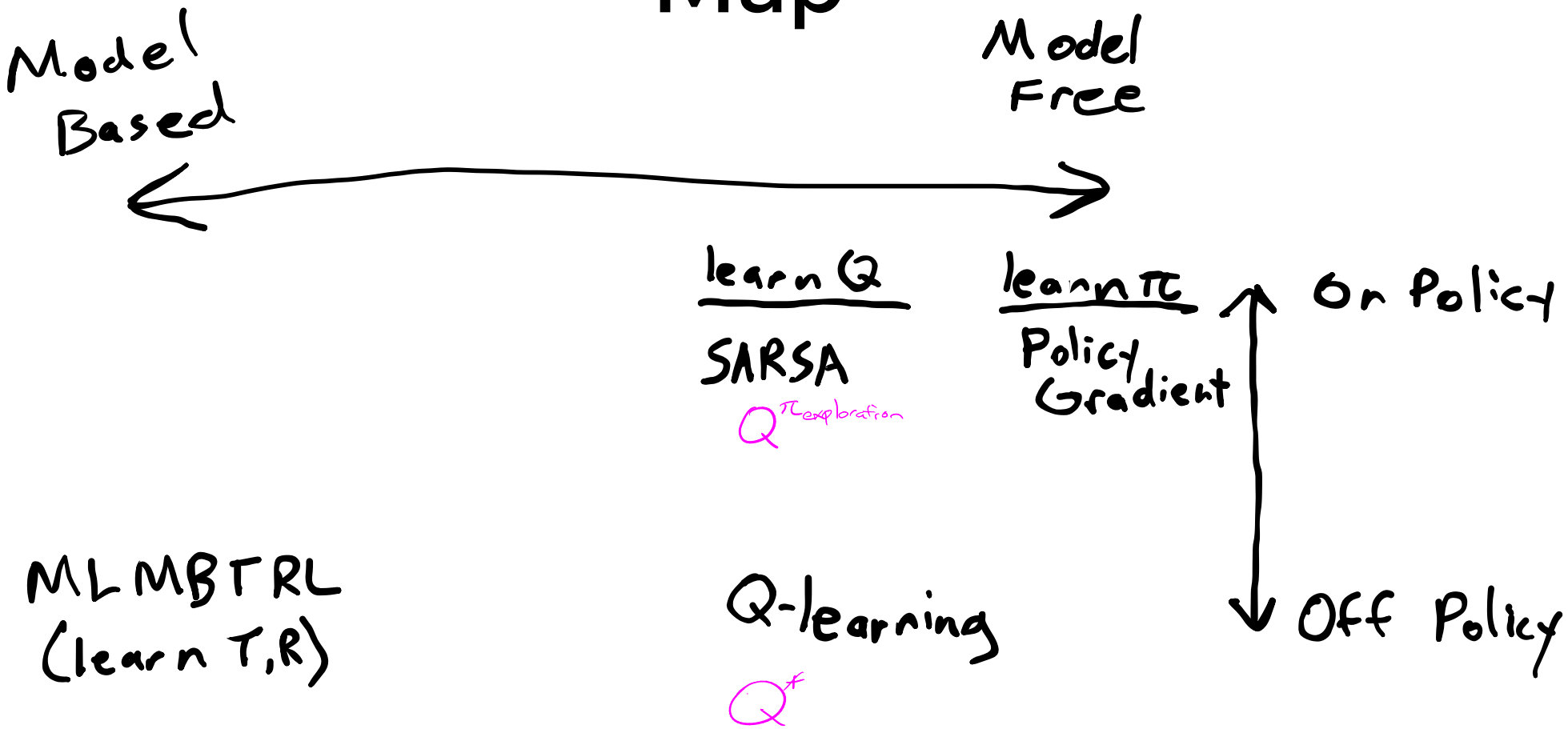
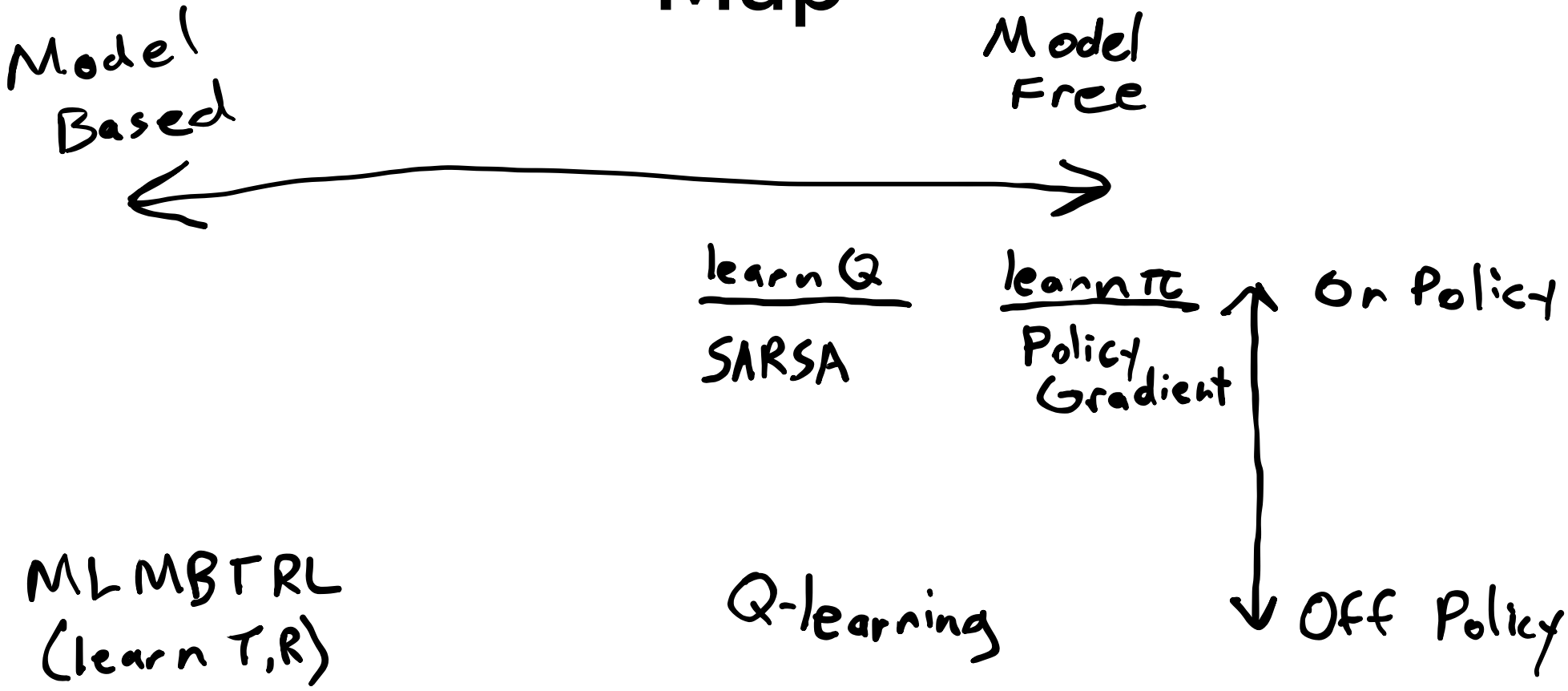

Model Based
Model Free
learn Q
learn π
SARSA
Policy Gradient
On Policy
Off Policy
ML MBTRL (learn T,R)
Q-learning

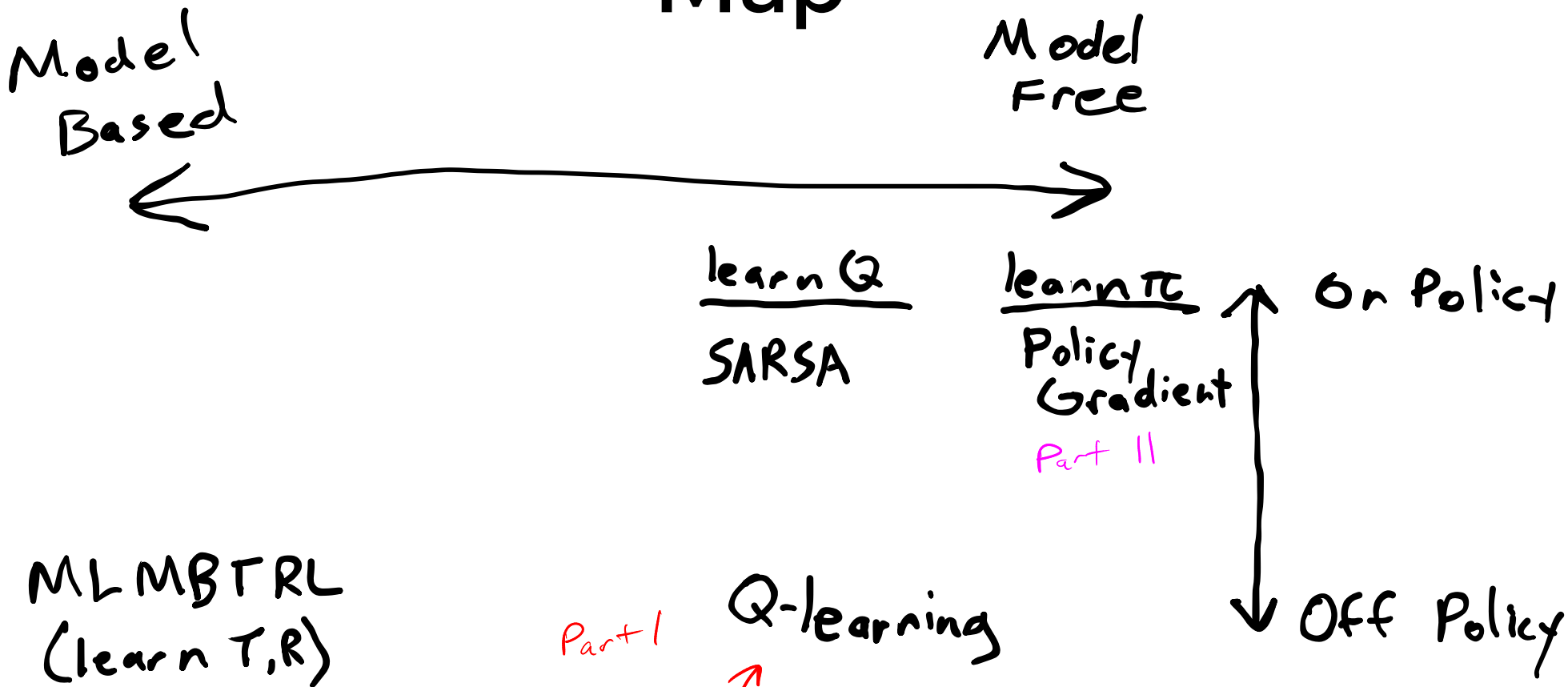
Map



Challenges:

1. Exploration vs Exploitation
2. Credit Assignment
3. Generalization

Map



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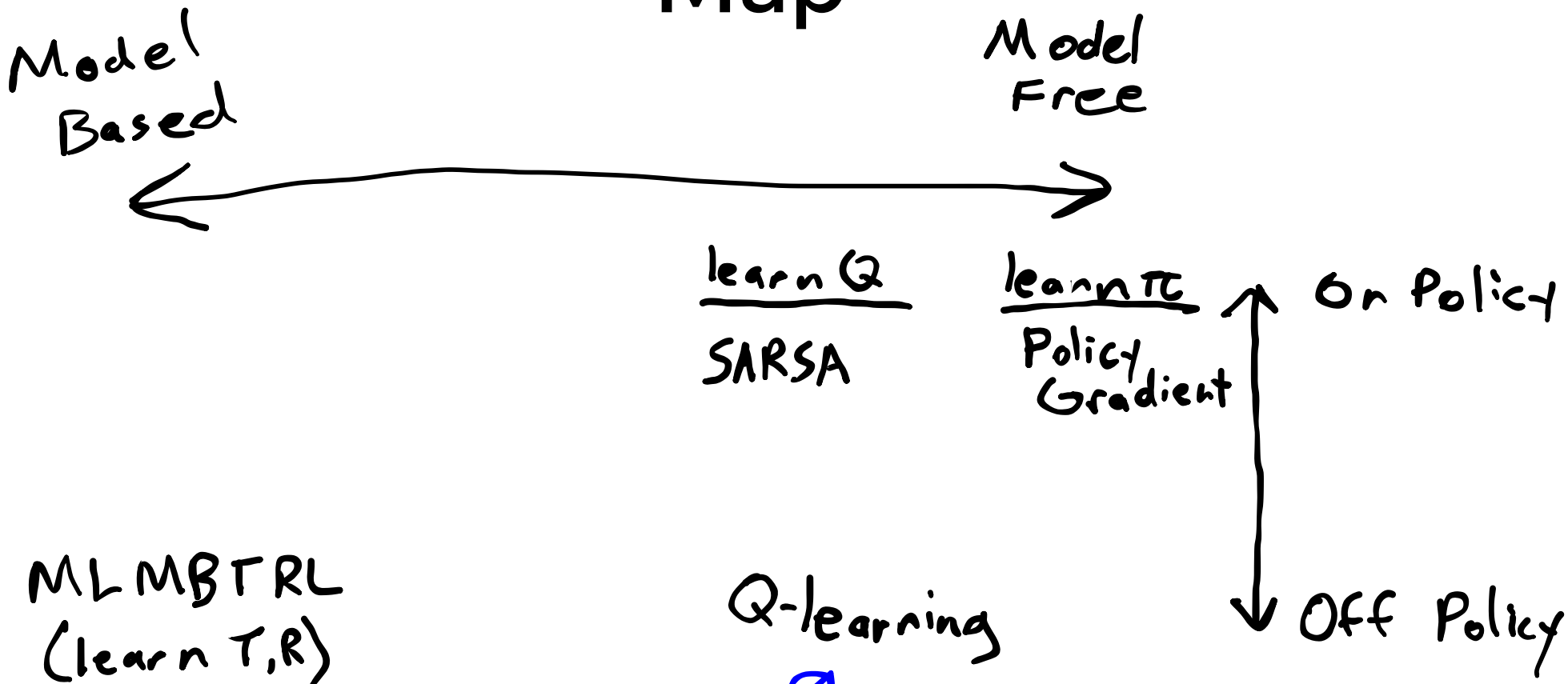

Challenges:

1. Exploration vs Exploitation
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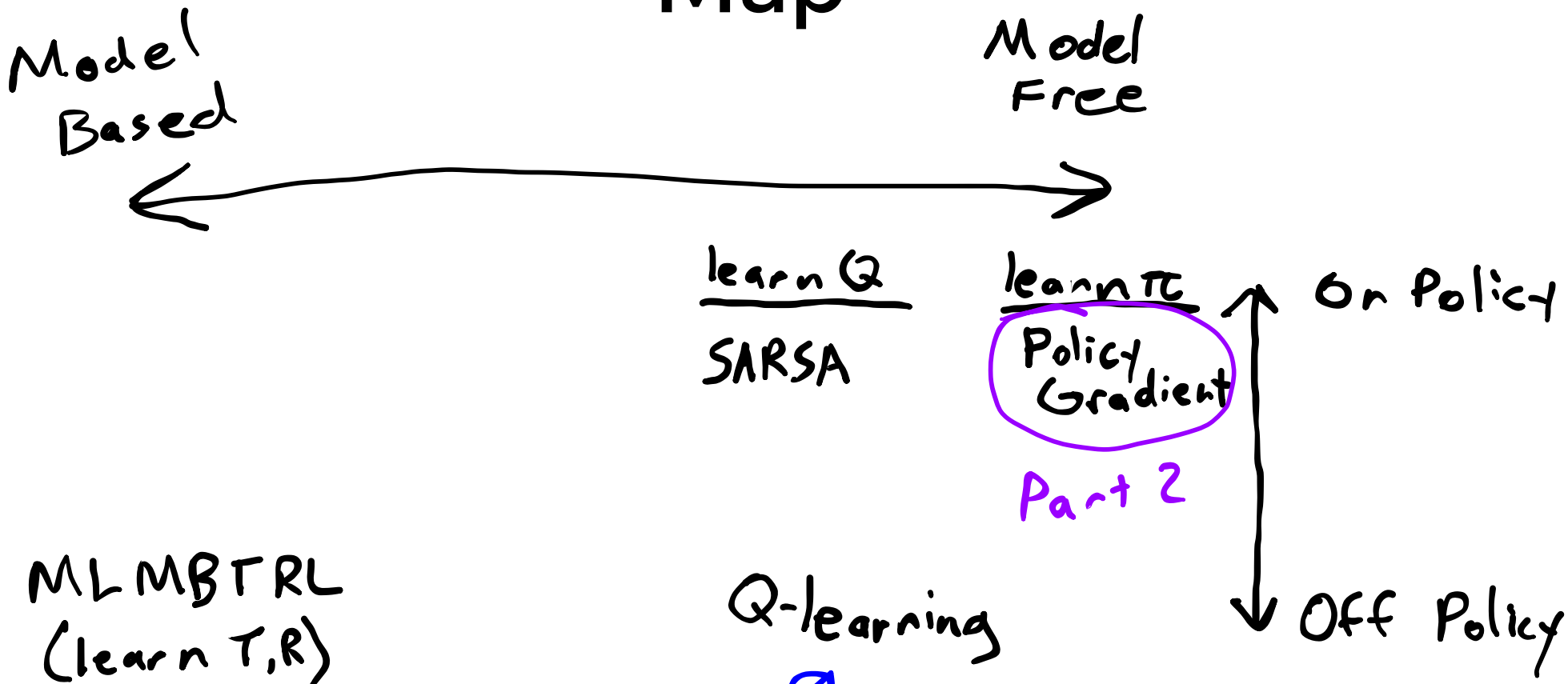
Last Time: Neural Networks

Map



Challenges:

1. Exploration vs Exploitation
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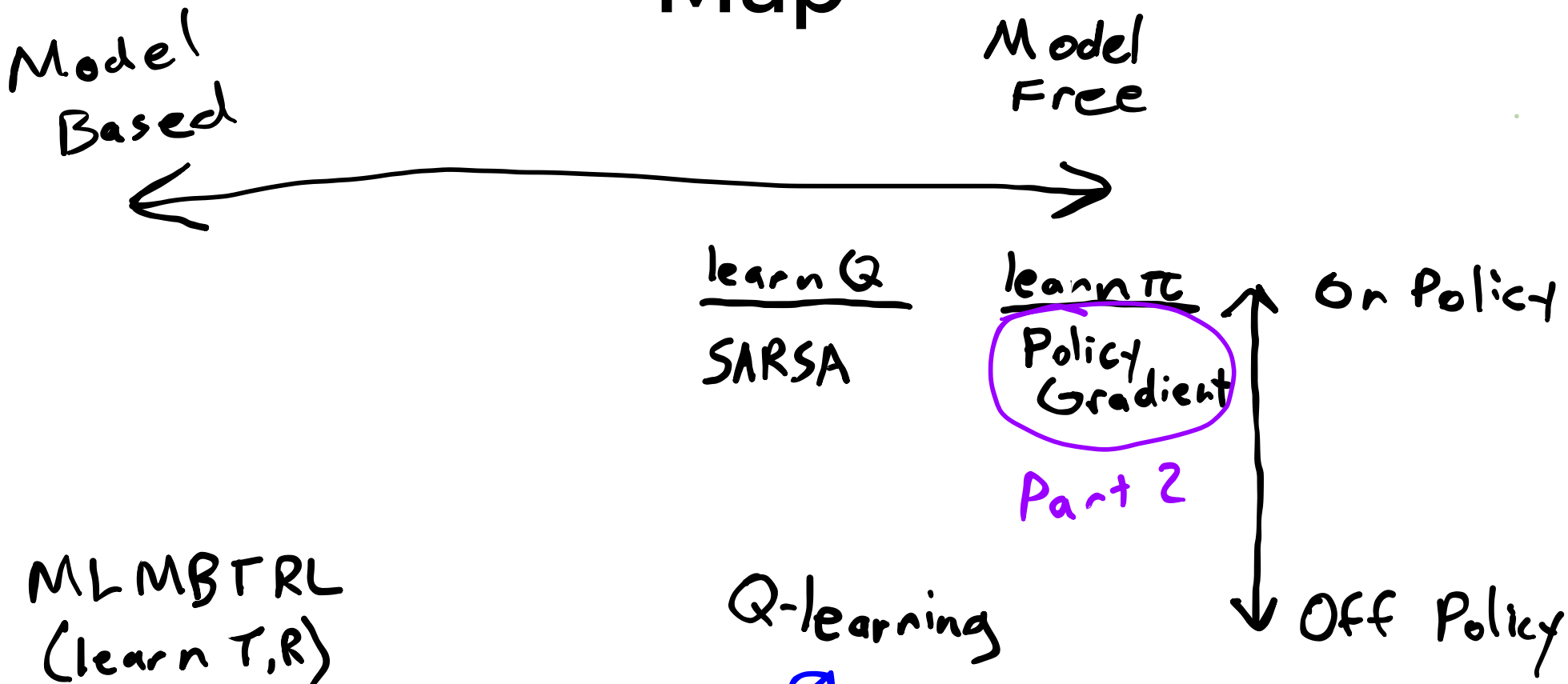

Challenges:

1. Exploration vs Exploitation
2. Credit Assignment
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Part I
DQN

Q-Learning with Neural Networks

Q-Learning with Neural Networks

Q-Learning:

$$Q(s,a) \leftarrow Q(s,a) + \alpha \left(r + \gamma \max_{a'} Q(s',a') - Q(s,a)\right)$$

Q-Learning with Neural Networks

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Neural Networks

$$\theta^* = \arg\min_{\theta} \sum_{(x,y)\in\mathcal{D}} l(f_\theta(x), y)$$

$$\theta \leftarrow \theta - \alpha \nabla_\theta l(f_\theta(x), y)$$

Q-Learning with Neural Networks

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Deep Q learning:

- Approximate Q with Q_θ

Q-Learning with Neural Networks

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- What should (x, y) be?

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Deep Q learning:

- Approximate Q with Q_θ
- What should (x, y) be?
- What should l be?

Want

$$r + \gamma \max_{a'} Q_\theta(s',a') = Q_\theta(s,a)$$

Loss

$$l(s,a,r,s') = \left(r + \gamma \max_{a'} Q_\theta(s',a') - Q_\theta(s,a)\right)^2$$

Q-Learning with Neural Networks

Q-Learning:

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Deep Q learning:

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Candidate Algorithm:

loop

$$a \leftarrow \text{argmax}\, Q(s,a) \text{ w.p. } 1-\epsilon, \quad \text{rand}(A) \text{ o.w.}$$

$$r \leftarrow \text{act!}(\text{env}, a)$$

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$$\theta \leftarrow \theta - \alpha \nabla_\theta \left(r + \gamma \max_{a'} Q_\theta(s',a') - Q_\theta(s,a)\right)^2$$

$$s \leftarrow s'$$

$\nabla_\theta\, l(s,a,r,s')$

DQN: The Atari Benchmark

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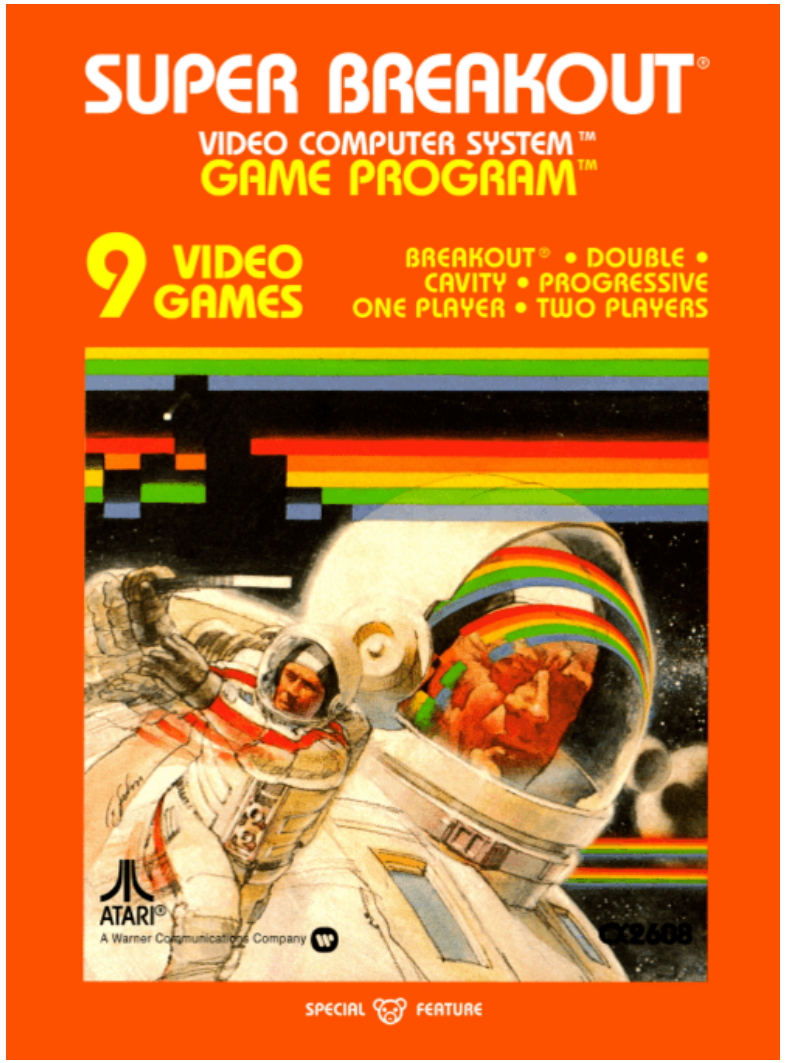
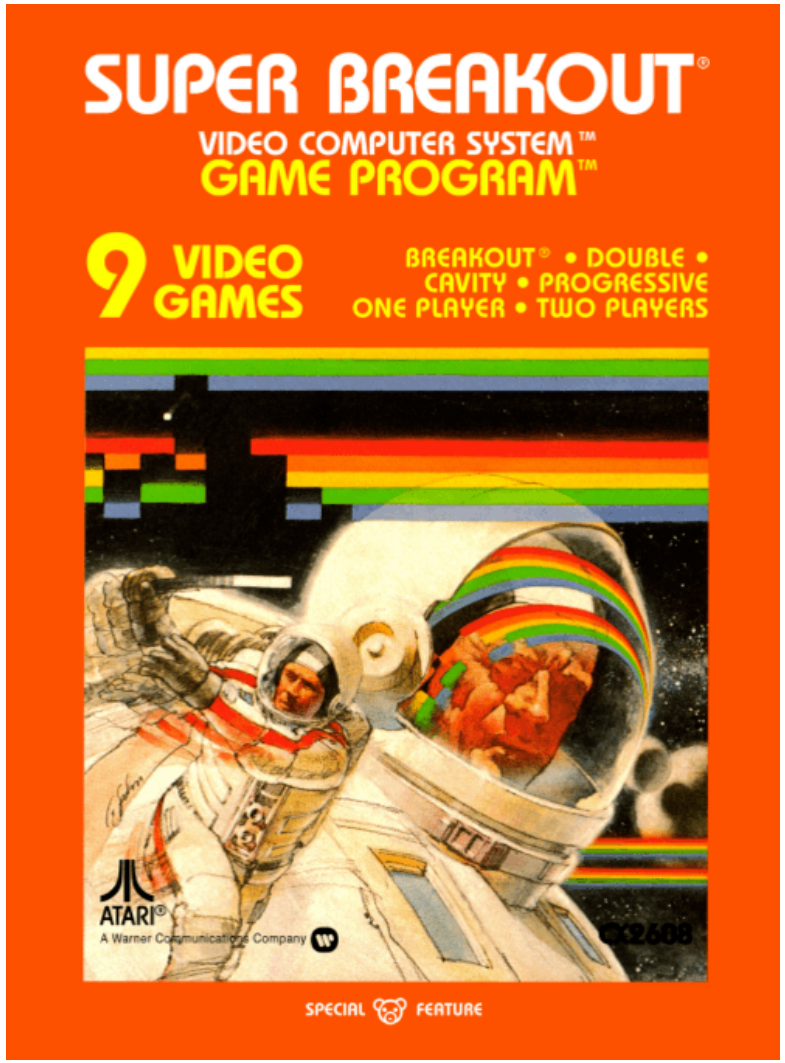


DQN: The Atari Benchmark



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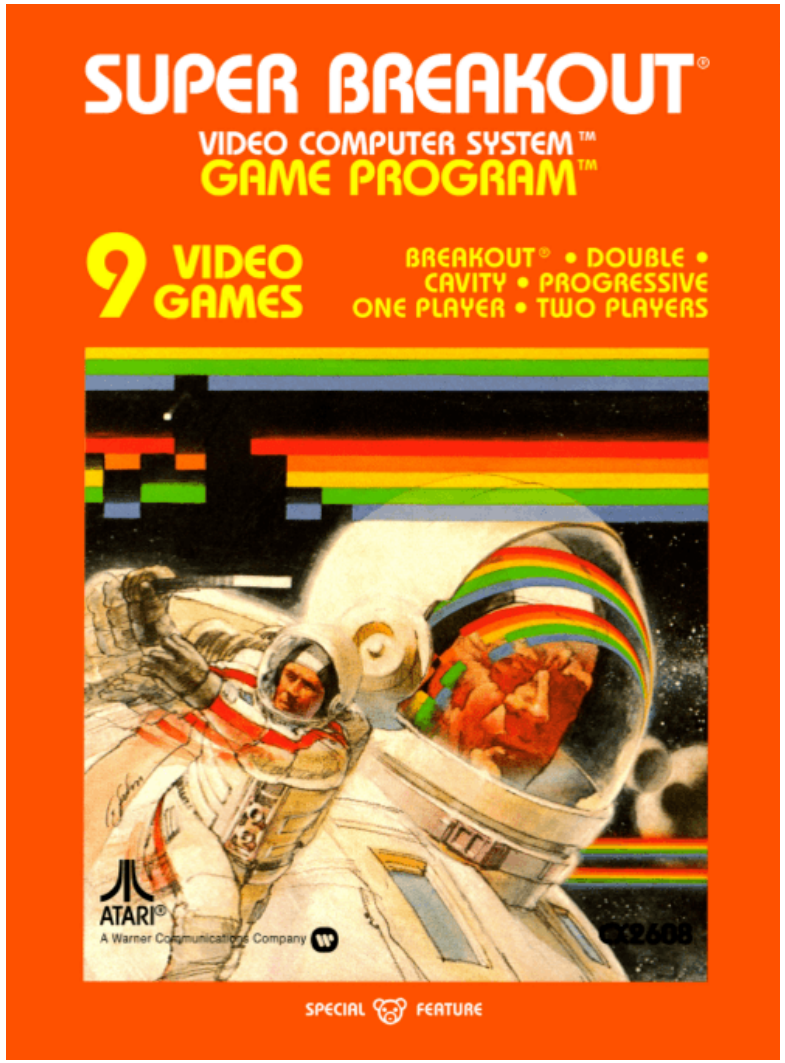

DQN: Problems with Naive Approach

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Problems:

1. Samples Highly Correlated

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1. Samples Highly Correlated
2. Size-1 batches

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Problems:

1. Samples Highly Correlated
2. Size-1 batches
3. Moving target

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Problems:

1. Samples Highly Correlated
2. Size-1 batches
3. Moving target

} data buffer / experience replay

} periodically freeze target

DQN

DQN

Q Network Structure:

DQN

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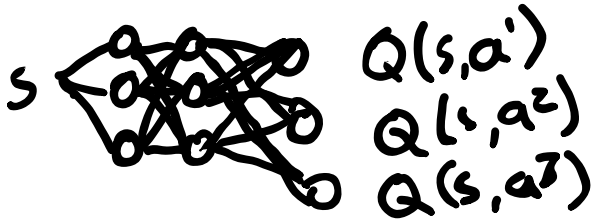
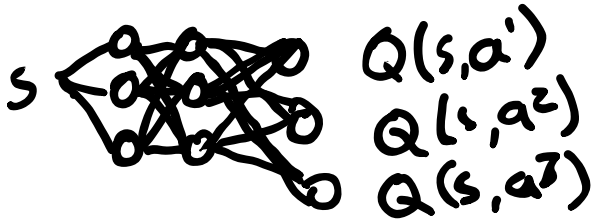


DQN

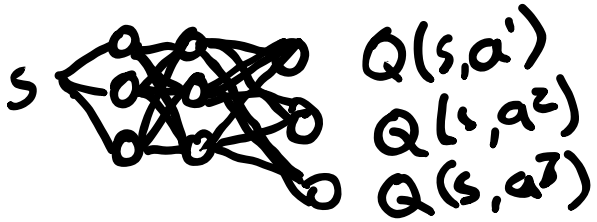
Q Network Structure:



Experience Tuple: (s, a, r, s')

DQN

Q Network Structure:

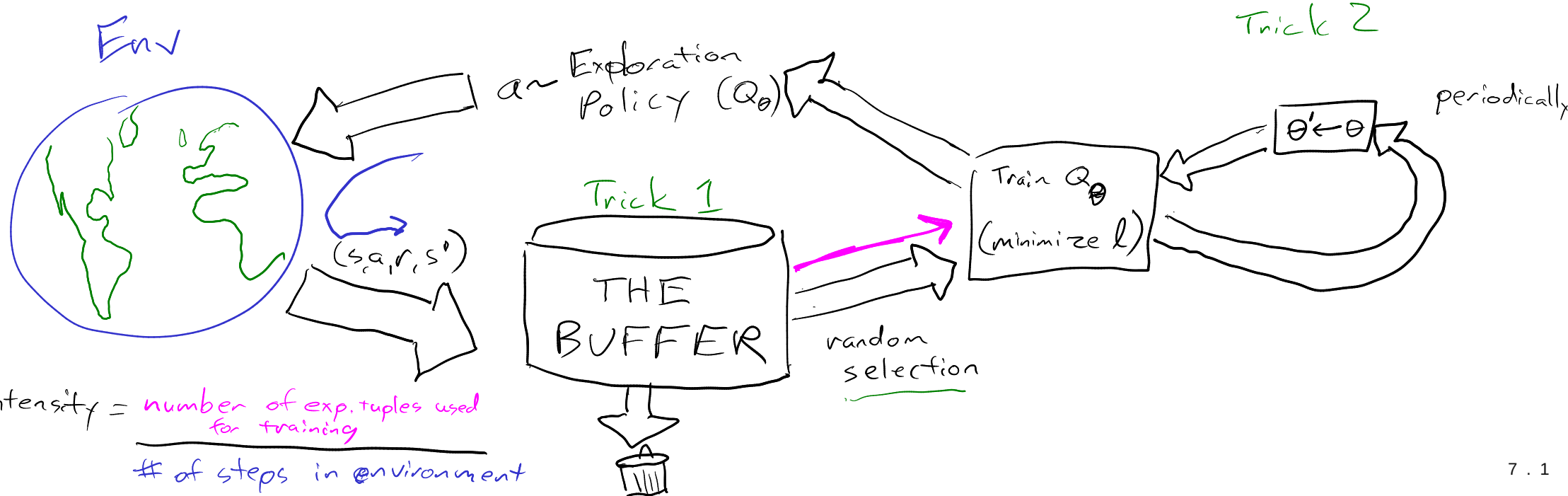


Experience Tuple: (s, a, r, s')

Loss:

$$l(s,a,r,s') = \left(r + \gamma \max_{a'} Q_{\theta'}(s',a') - Q_{\theta}(s,a)\right)^2$$

Double Q: $Q_{\theta_1}(s', \text{argmax}_{a'} Q_{\theta_2}(s',a'))$



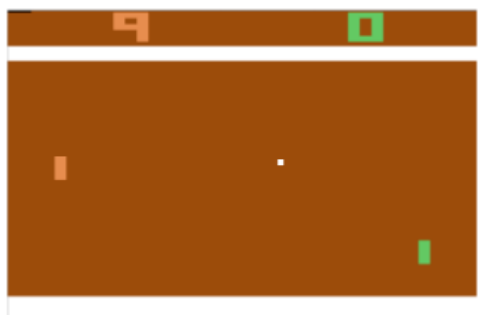
Training Intensity = $\frac{\text{number of exp. tuples used for training}}{\text{\# of steps in environment}}$

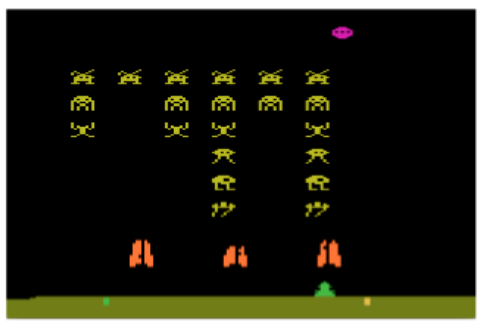
Playing Atari with Deep Reinforcement Learning

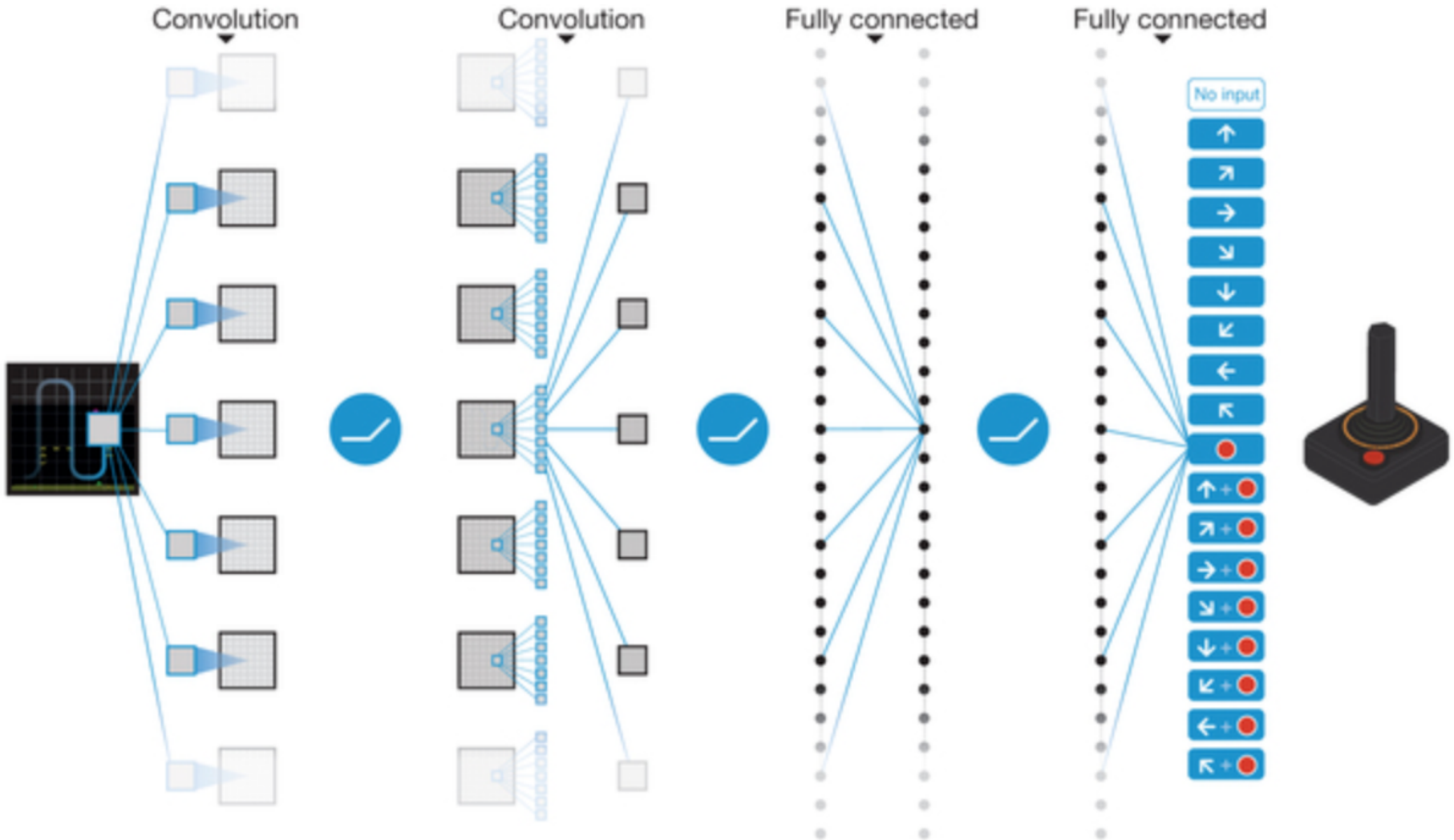
Volodymyr Mnih **Koray Kavukcuoglu** **David Silver** **Alex Graves** **Ioannis Antonoglou**

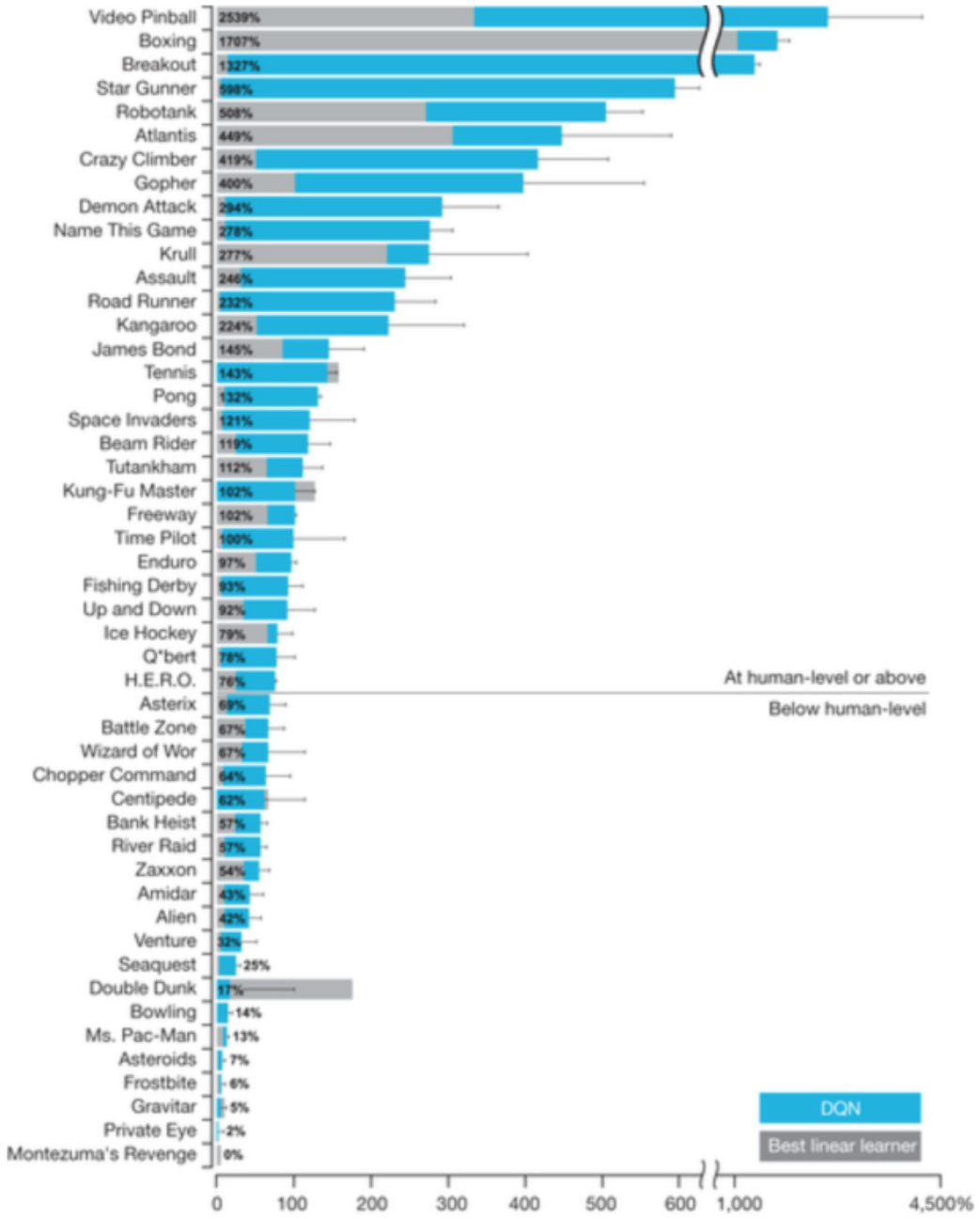
Daan Wierstra **Martin Riedmiller**

DeepMind Technologies

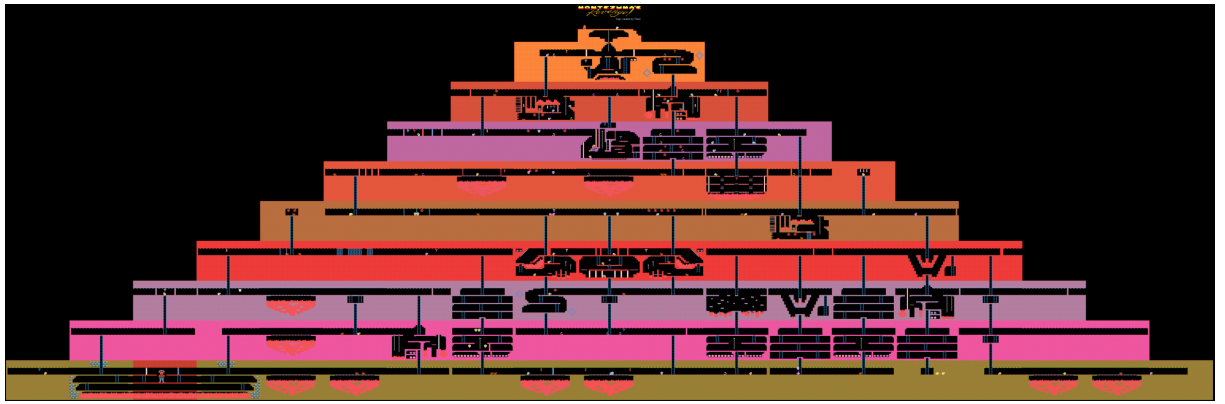






Convolution
Convolution
Fully connected
Fully connected
No input



https://www.youtube.com/watch?v=SuZVyOlgVek



Rainbow

Rainbow

- Double Q Learning

Rainbow

- Double Q Learning
- Prioritized Replay
  (priority proportional to last TD error)

Rainbow

- Double Q Learning
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  (priority proportional to last TD error)
- Dueling networks
  Value network + advantage network
  $Q(s, a) = V(s) + A(s, a)$

Rainbow

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  $Q(s,a) = V(s) + A(s,a)$
- Multi-step learning
  $(r_t + \gamma r_{t+1} + \ldots + \gamma^{n-1} r_{t+n-1} + \gamma \max Q_\theta(s_{t+n}, a') - Q_\theta(s_t, a_t))^2$

Rainbow

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  predict an entire distribution of values instead of just Q

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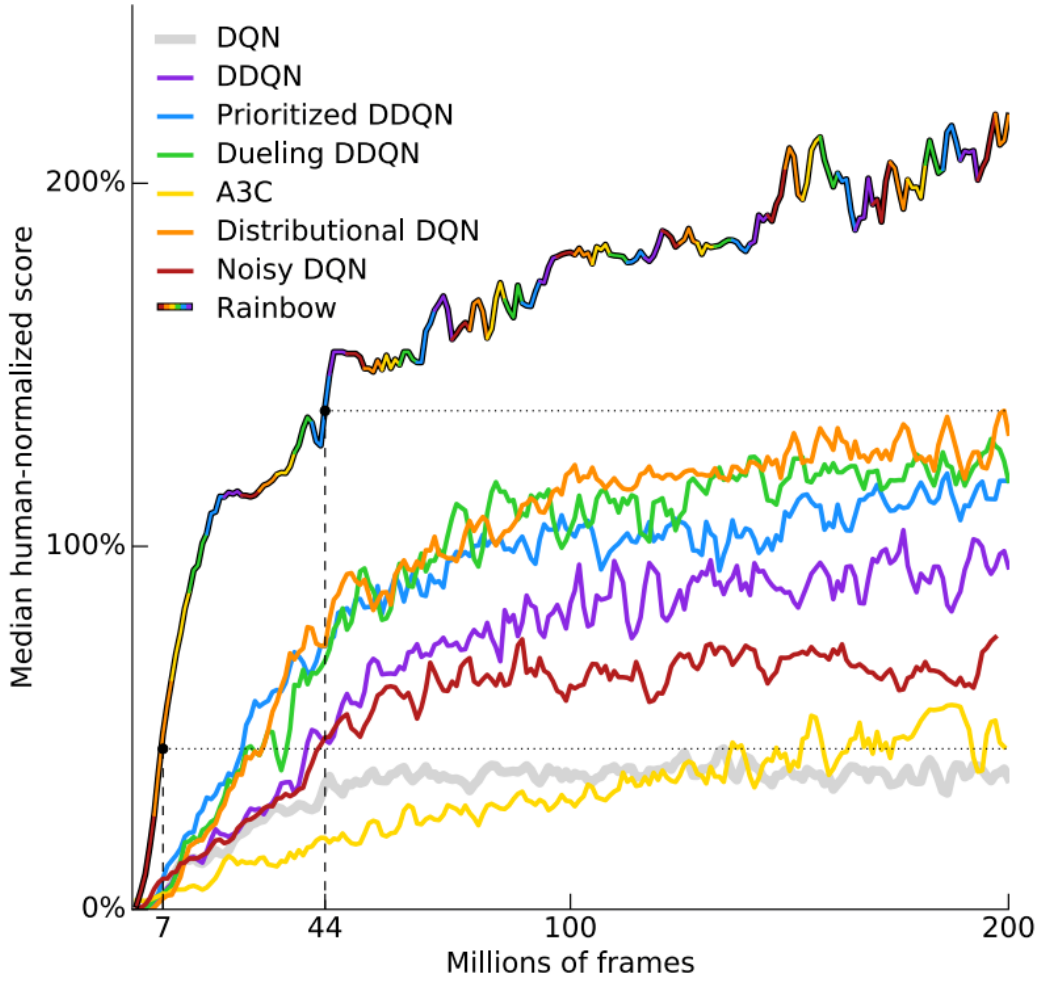
- Noisy Nets

Rainbow



- Double Q Learning
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Actual Learning Curves

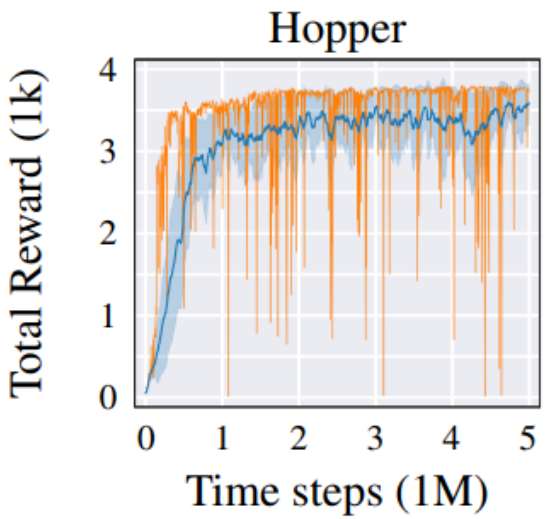

Figure 24: Standard presentation Single seed
The training curve as commonly presented in RL papers and the training curve of a single seed . Both curves are from the TD3 algorithm, trained for 5M time steps. The standard presentation is to evaluate every N_{freq} steps, average scores over N_{episodes} evaluation episodes and N_{seeds} seeds, then to smooth the curve by averaging over a window of N_{window} evaluations. (In our case this corresponds to $N_{\text{freq}} = 5000$, $N_{\text{episodes}} = 10$, $N_{\text{seeds}} = 10$, $N_{\text{window}} = 10$). The learning curve of a single seed has no smoothing over seeds or evaluations ($N_{\text{seeds}} = 1$, $N_{\text{window}} = 1$). By averaging over many seeds and evaluations, the training curves in RL can appear deceptively smooth and stable.

Paper: For SALE: State-Action Representation Learning for Deep Reinforcement Learning

Part II
Improved Policy Gradients

Policy Gradient

$$\widehat{\nabla_\theta U(\theta)} = \sum_{k=0}^{d} \nabla_\theta \log \pi_\theta(a_k \mid s_k) \gamma^k (r_{k,\text{to-go}} - r_{\text{base}}(s_k))$$

Policy Gradient

$$\widehat{\nabla_\theta U(\theta)} = \sum_{k=0}^{d} \nabla_\theta \log \pi_\theta(a_k \mid s_k) \gamma^k (r_{k,\text{to-go}} - r_{\text{base}}(s_k))$$

$$\uparrow$$

$$\hat{U}^\pi(s_k)$$

Policy Gradient

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$$\uparrow \qquad\qquad \uparrow$$

$$\hat{Q}^\pi(s_k, a_k) \qquad \hat{U}^\pi(s_k)$$

Policy Gradient

$$\widehat{\nabla_\theta U(\theta)} = \sum_{k=0}^{d} \nabla_\theta \log \pi_\theta(a_k \mid s_k) \gamma^k (r_{k,\text{to-go}} - r_{\text{base}}(s_k))$$

$$\uparrow \qquad\qquad \uparrow$$

$$\hat{Q}^\pi(s_k, a_k) \qquad \hat{U}^\pi(s_k)$$

Advantage Function:

$$A(s,a) \equiv Q(s,a) - U(s)$$

$$A^*(s,a) = Q^*(s,a) - U^*(s)$$

$$A^\pi(s,a) = Q^\pi(s,a) - U^\pi(s)$$

Policy Gradient

$$\widehat{\nabla_\theta U(\theta)} = \sum_{k=0}^{d} \nabla_\theta \log \pi_\theta(a_k \mid s_k) \gamma^k (r_{k,\text{to-go}} - r_{\text{base}}(s_k))$$

$$\uparrow \qquad \uparrow$$

$$\hat{Q}^\pi(s_k, a_k) \qquad \hat{U}^\pi(s_k)$$

Advantage Function:

$A(s, a) \equiv Q(s, a) - U(s)$

loop:

$\tau \leftarrow \text{simulate\_episode}(\pi_\theta)$

$\widehat{\nabla_\theta U(\theta)} \leftarrow \sum_{k=0}^{d} \nabla_\theta \log \pi_\theta(a_k \mid s_k) \gamma^k \hat{A}_k$

$\theta \leftarrow \theta + \alpha \widehat{\nabla_\theta U(\theta)}$

Policy Gradient

$$\widehat{\nabla_\theta U(\theta)} = \sum_{k=0}^{d} \nabla_\theta \log \pi_\theta(a_k \mid s_k) \gamma^k (r_{k,\text{to-go}} - r_{\text{base}}(s_k))$$

$$\uparrow \qquad \uparrow$$

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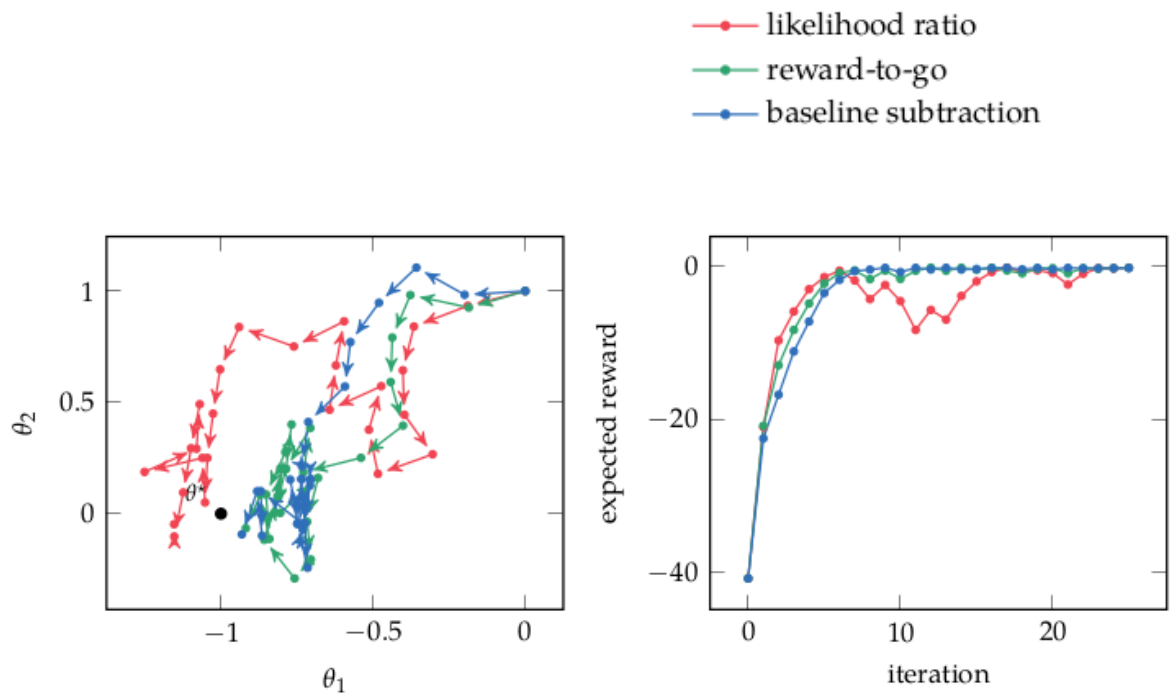

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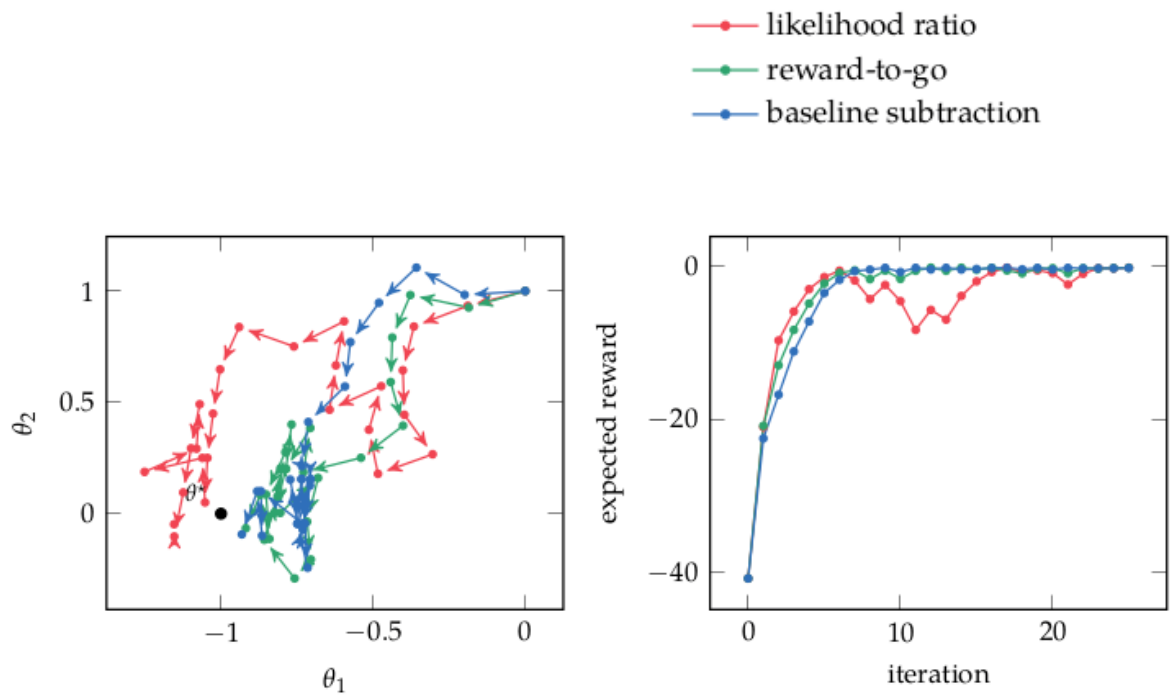

Some problems:

- $\widehat{\nabla_\theta U(\theta)}$ has high variance
- The gradient is not a good step direction
- Fixed α steps are too big or small

Trust Region Optimization

$$\widehat{\nabla U}(\theta) = \sum_{k=0}^{d} \nabla_\theta \log \pi_\theta(a_k \mid \boldsymbol{s}_k) \gamma^k \, \hat{A}_k$$

$$\theta' = \theta + \alpha \widehat{\nabla U}(\theta)$$

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$$\begin{aligned} \underset{\theta'}{\text{maximize}} \quad & U(\theta') \approx U(\theta) + \nabla U(\theta)^\top (\theta' - \theta) \\ \text{subject to} \quad & g(\theta, \theta') \leq \epsilon \end{aligned}$$

Trust Region Optimization

$$\widehat{\nabla U}(\theta) = \sum_{k=0}^{d} \nabla_\theta \log \pi_\theta(a_k \mid s_k) \gamma^k \hat{A}_k$$

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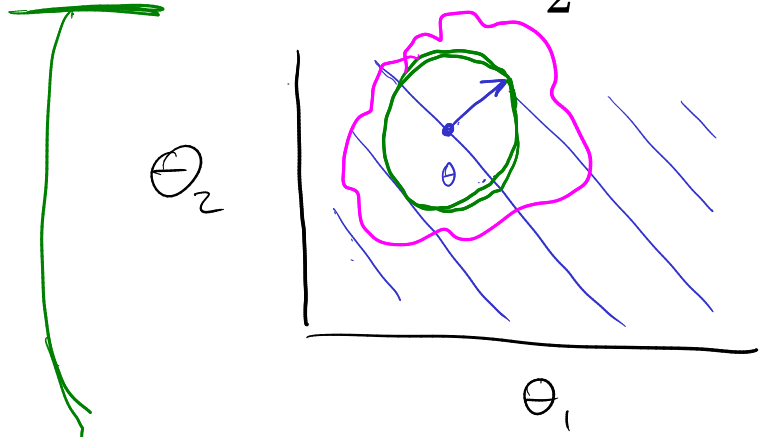
$$\text{subject to} \qquad g(\theta, \theta') \leq \epsilon$$

Example:

$$g(\theta, \theta') = \|\theta - \theta'\|_2^2 = \frac{1}{2}(\theta' - \theta)^\top (\theta' - \theta)$$



$$\theta' = \theta + \mathbf{u}\sqrt{\frac{2\epsilon}{\mathbf{u}^\top \mathbf{u}}} = \theta + \sqrt{2\epsilon}\frac{\mathbf{u}}{\|\mathbf{u}\|}$$

$$\mathbf{u} = \nabla U(\theta)$$

Natural Gradient

$$g(\theta, \theta') = D_{\mathrm{KL}}\left(p(\cdot \mid \theta) \,\middle|\middle|\, p(\cdot \mid \theta')\right) \leq \epsilon$$

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↳ Fischer Information Matrix

$$\begin{aligned} \mathbf{F}_\theta &= \int p(\tau \mid \theta) \nabla \log p(\tau \mid \theta) \nabla \log p(\tau \mid \theta)^\top \, \mathrm{d}\tau \\ &= \mathbb{E}_\tau \Big[\nabla \log p(\tau \mid \theta) \nabla \log p(\tau \mid \theta)^\top \Big] \end{aligned}$$

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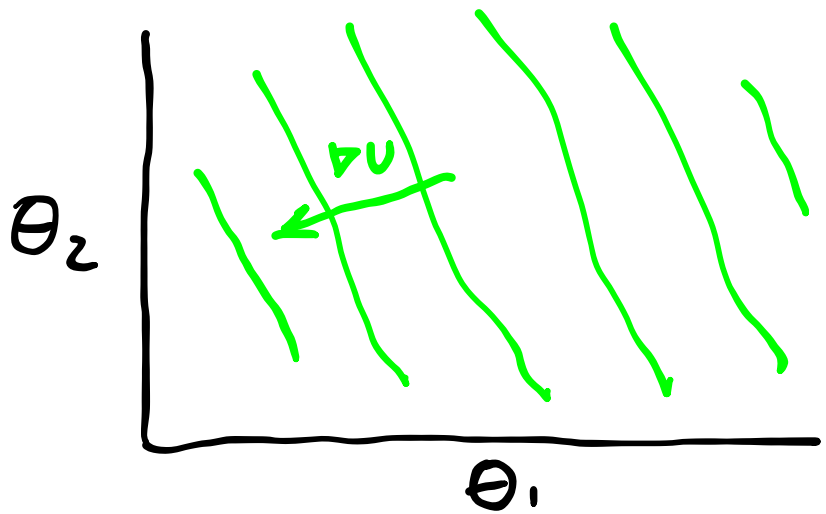

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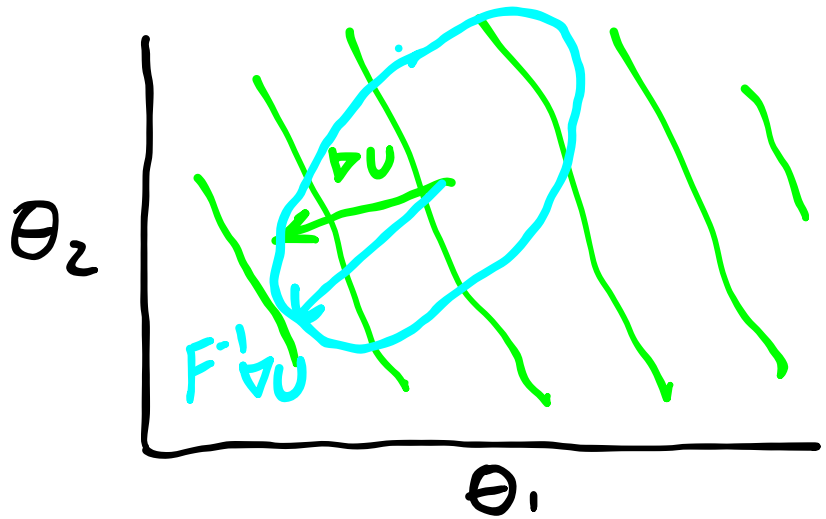

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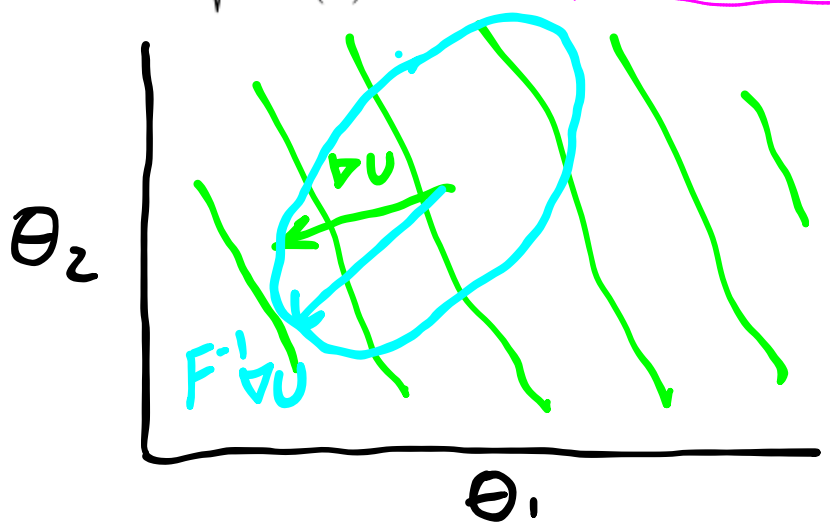

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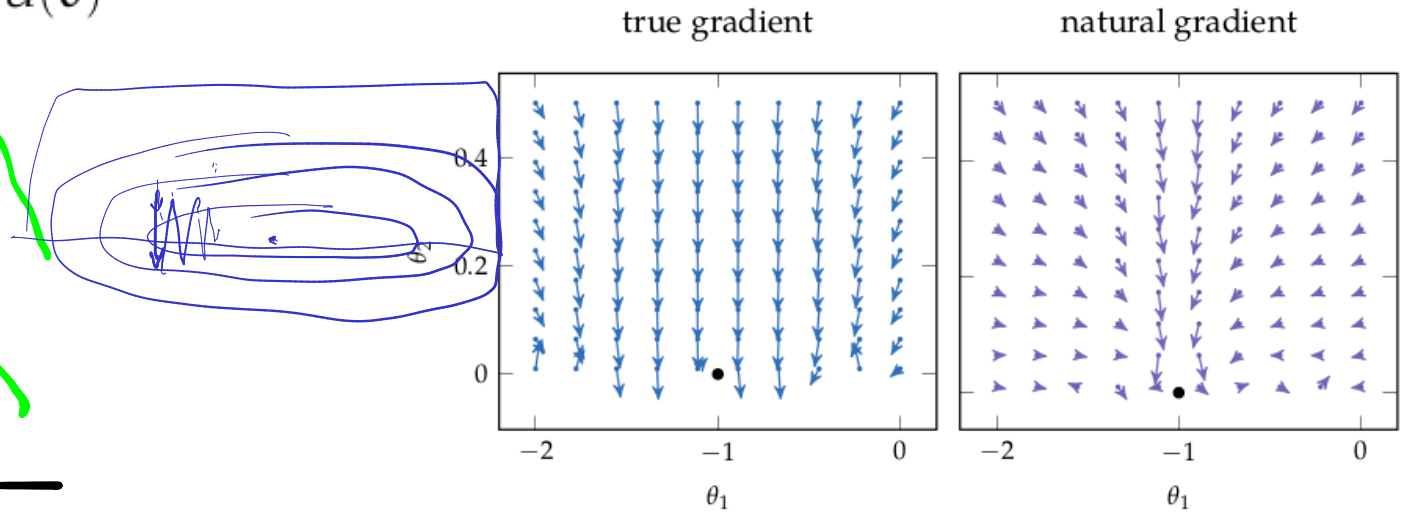

Conservative Policy Iteration Objective

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Discounted Visitation Distribution

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$$\underset{\tau \sim \pi_\theta}{E}\left[\sum_{t=0}^{\infty} \gamma^t R(s_t, a_t)\right]$$

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$$\underset{\tau \sim \pi_\theta}{E}\left[\sum_{t=0}^{\infty} \gamma^t R(s_t, a_t)\right] = \underset{\substack{s \sim b_{\gamma,\theta} \\ a \sim \pi_\theta(\cdot|s)}}{E}[R(s,a)]$$

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$$b_{\gamma,\theta}(s) \propto P(s_1 = s) + \gamma P(s_2 = s) + \gamma^2 P(s_3 = s) + \dots$$

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Policy Advantage

$$f(\theta, \theta') = \underset{\substack{s \sim b_{\gamma,\theta} \\ a \sim \pi_{\theta'}(\cdot|s)}}{E}\left[A^{\pi_\theta}(s,a)\right]$$

$$= \underset{\substack{s \sim b_{\gamma,\theta} \\ a \sim \pi_\theta(\cdot|s)}}{E}\left[\frac{\pi_{\theta'}(a|s)}{\pi_\theta(a|s)} A^{\pi_\theta}(s,a)\right]$$

"Approximately Optimal Approximate RL"
Kakade & Langford

Trust Region Policy Optimization (TRPO)

Natural Gradient Approximation + Conservative Policy Iteration Objective

Update Step:

$$\underset{\theta'}{\text{maximize}} \quad f(\theta, \theta') = \underset{\substack{s \sim b_{\gamma,\theta} \\ a \sim \pi_\theta(\cdot|s)}}{E} \left[\frac{\pi_{\theta'}(a \mid s)}{\pi_\theta(a \mid s)} A^{\pi_\theta}(s, a) \right]$$

$$\text{subject to} \quad g(\theta, \theta') = \underset{s \sim b_{\gamma,\theta}}{\mathbb{E}} \left[D_{\mathrm{KL}}(\pi_\theta(\cdot \mid s) \parallel \pi_{\theta'}(\cdot \mid s)) \right] \leq \epsilon$$

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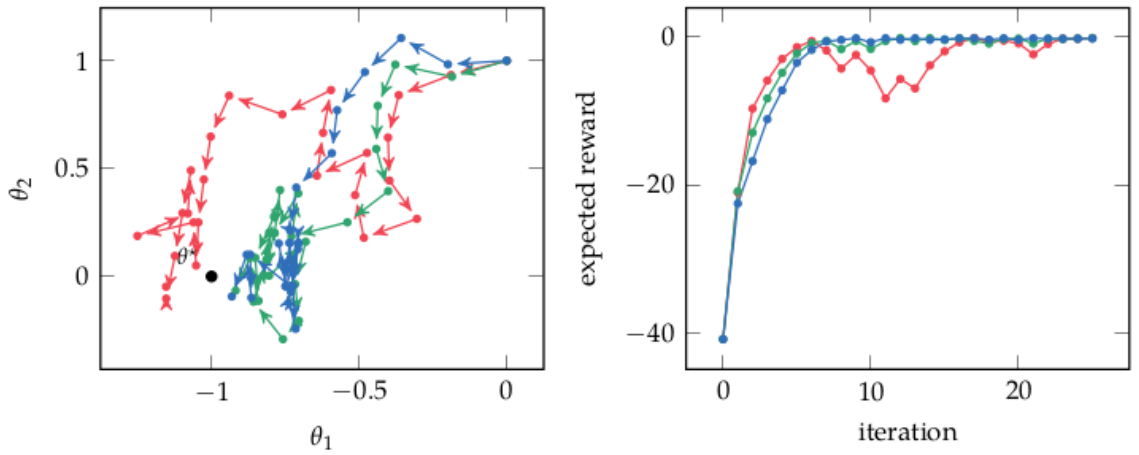

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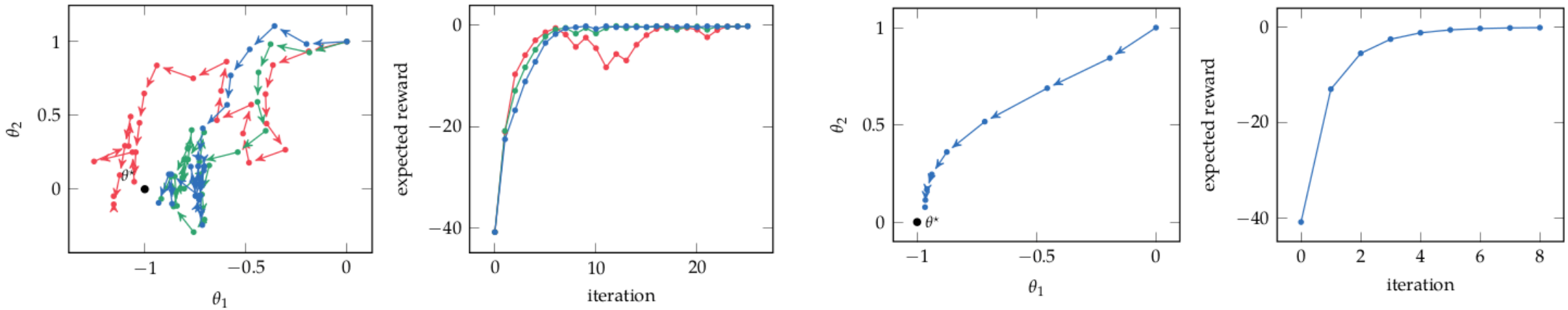

Proximal Policy Optimization (PPO)

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Clipped Surrogate Objective:

Proximal Policy Optimization (PPO)

Clipped Surrogate Objective:

$$\underset{\substack{s\sim b_{\gamma,\theta}\\ a\sim\pi_\theta(\cdot|s)}}{E}\left[\min\left(\frac{\pi_{\theta'}(a\mid s)}{\pi_\theta(a\mid s)}A_\theta(s,a),\operatorname{clamp}\left(\frac{\pi_{\theta'}(a\mid s)}{\pi_\theta(a\mid s)},1-\epsilon,1+\epsilon\right)A_\theta(s,a)\right)\right]$$

Proximal Policy Optimization (PPO)

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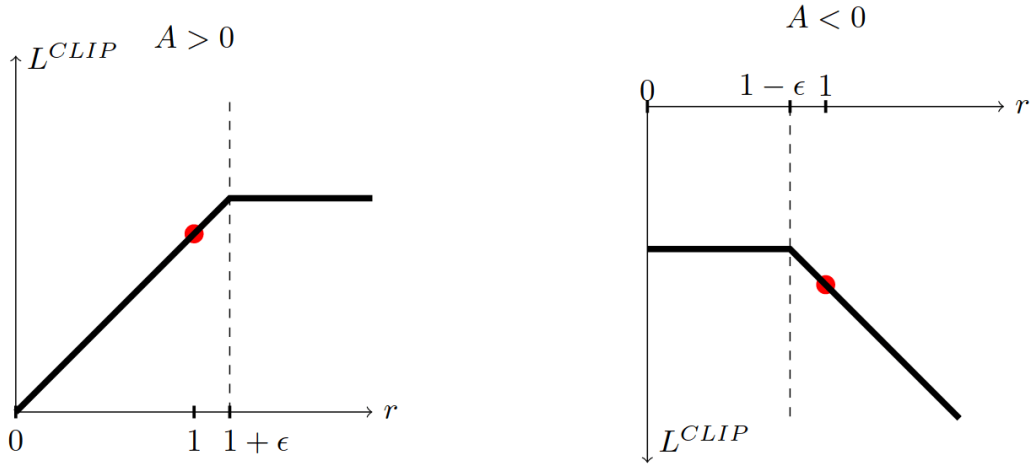

$$r=\frac{\pi_{\theta'}(a\mid s)}{\pi_\theta(a\mid s)}$$

Proximal Policy Optimization (PPO)

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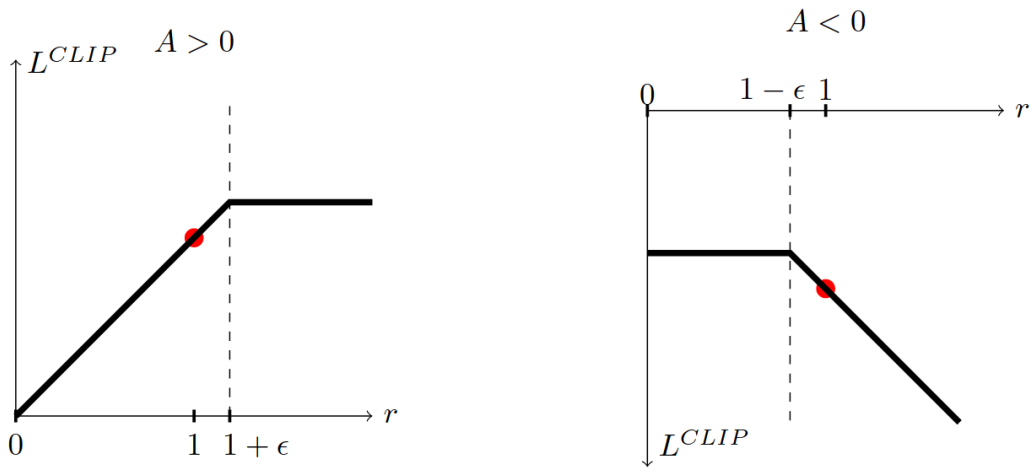

$$r=\frac{\pi_{\theta'}(a\mid s)}{\pi_\theta(a\mid s)}$$

Each step: take several gradient steps with trajectories from π_θ

(No need for constraint because it will stop when it hits the clip)

PPO Impact

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Cited 29262 times (as of Fall 2025)

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(avg ~10x per day)

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But...

Implementation Matters

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Implementation Matters in Deep Policy Gradients: A Case Study on PPO and TRPO

Logan Engstrom, Andrew Ilyas, Shibani Santurkar, Dimitris Tsipras, Firdaus Janoos, Larry Rudolph, and Aleksander Madry; ICLR 2020

Implementation Details

- Value function clipping
- Reward scaling
- Orthogonal initialization and layer scaling
- Adam learning rate annealing
- Reward clipping
- Observation normalization
- Observation clipping
- Hyperbolic tan activations
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"PPO's marked improvement over TRPO (and even stochastic gradient descent) can be largely attributed to these optimizations."

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The 37 Implementation Details of Proximal Policy Optimization

Part III
Actor-Critic

Actor-Critic

Actor-Critic

Which should we learn? A, Q, or V?

Actor-Critic

Which should we learn? A, Q, or V?

$$\nabla U(\theta) = E_\tau \left[\sum_{k=0}^{d} \nabla_\theta \log \pi_\theta(a_k \mid s_k) \gamma^k (r_k + \gamma V_\phi(s_{k+1}) - V_\phi(s_k))) \right]$$

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temporal difference residual

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temporal difference residual

$$l(\phi) = E\left[\left(V_\phi(s) - V^{\pi_\theta}(s) \right)^2 \right]$$

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temporal difference residual

$$l(\phi) = E\left[\left(V_\phi(s) - V^{\pi_\theta}(s) \right)^2 \right]$$

↑ estimate with reward to go from sims

Generalized Advantage Estimation

Recap

Alpha Zero: Actor Critic with MCTS

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1. Use π_θ and U_ϕ in MCTS

Alpha Zero: Actor Critic with MCTS

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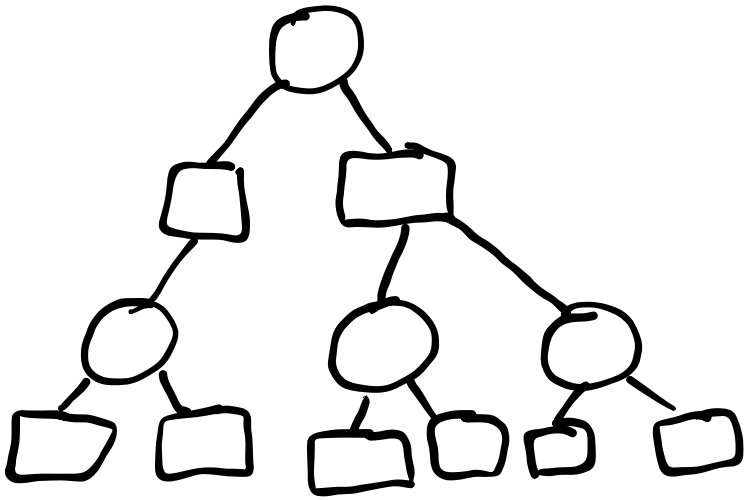

Alpha Zero: Actor Critic with MCTS

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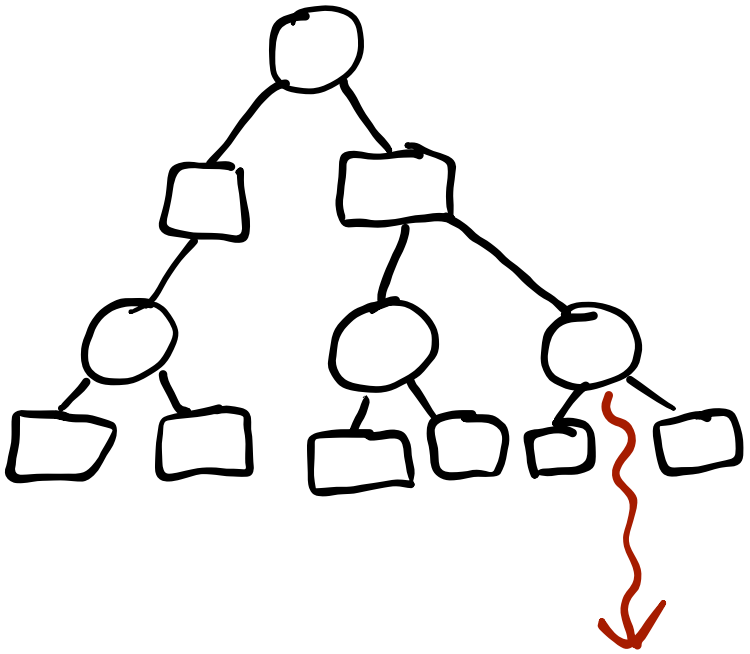

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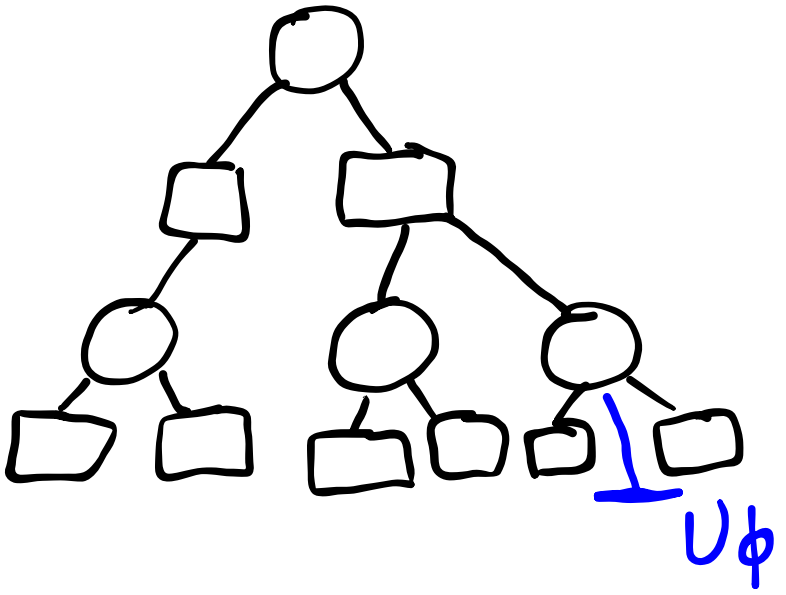

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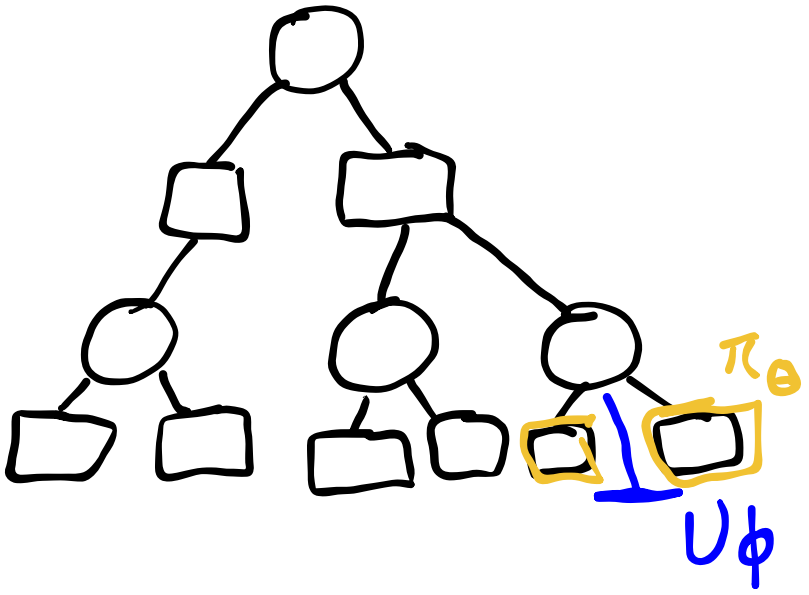
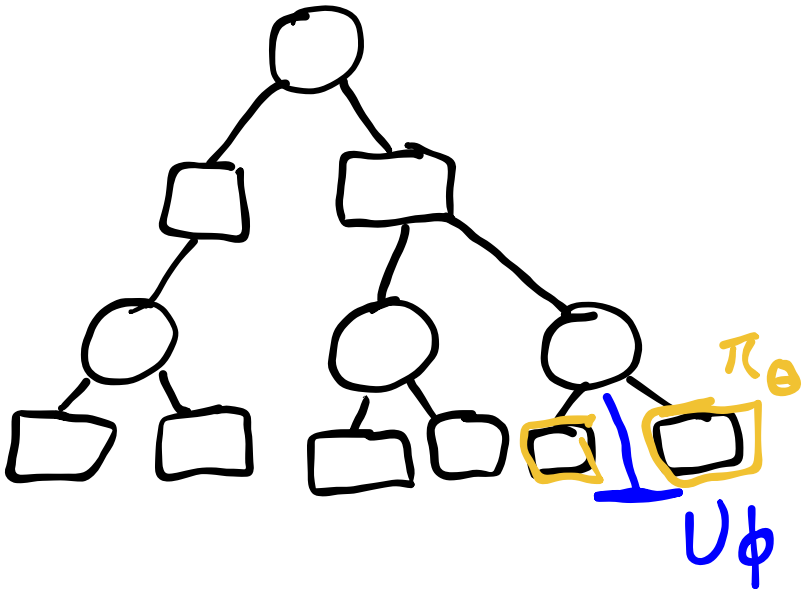


$$a = \arg\max_a Q(s,a) + c\pi_\theta(a \mid s)\frac{\sqrt{N(s)}}{1+N(s,a)}$$

Alpha Zero: Actor Critic with MCTS

1. Use π_θ and U_ϕ in MCTS
2. Learn π_θ and U_ϕ from tree



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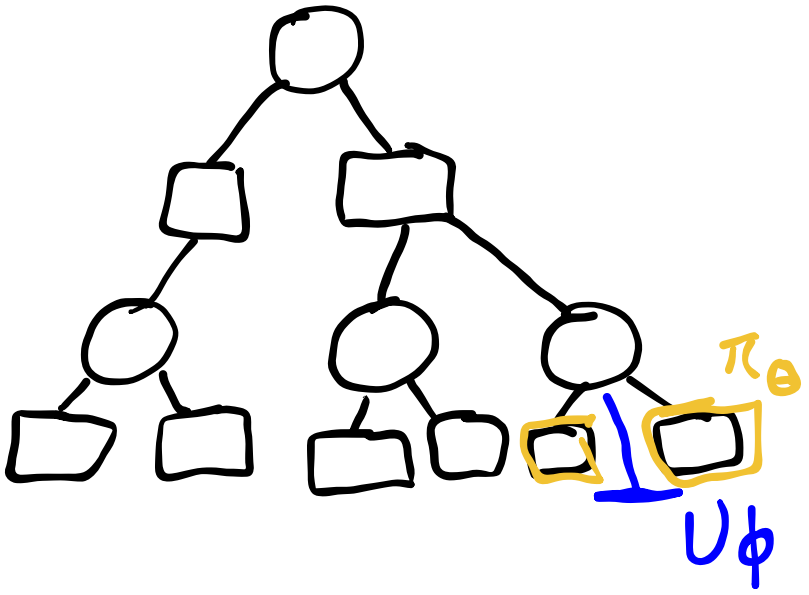
Alpha Zero: Actor Critic with MCTS

1. Use π_θ and U_ϕ in MCTS
2. Learn π_θ and U_ϕ from tree

$$\ell(\theta) = -\mathbb{E}_s\left[\sum_a \pi_{\text{MCTS}}(a \mid s) \log \pi_\theta(a \mid s)\right]$$

$$\pi_{\text{MCTS}}(a \mid s) \propto N(s,a)^\eta$$



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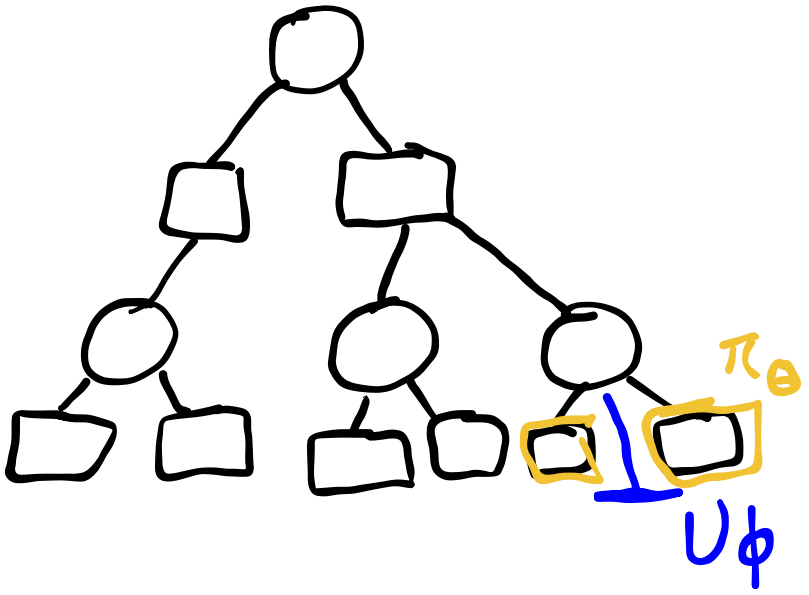
1. Use π_θ and U_ϕ in MCTS
2. Learn π_θ and U_ϕ from tree

$$\ell(\theta) = -\mathbb{E}_s\left[\sum_a \pi_{\text{MCTS}}(a \mid s) \log \pi_\theta(a \mid s)\right]$$

$$\pi_{\text{MCTS}}(a \mid s) \propto N(s,a)^\eta$$



$$\ell(\phi) = \frac{1}{2}\mathbb{E}_s\left[\left(U_\phi(s) - U_{\text{MCTS}}(s)\right)^2\right]$$

$$U_{\text{MCTS}}(s) = \max_a Q(s,a)$$

$$a = \arg\max_a Q(s,a) + c\pi_\theta(a \mid s)\frac{\sqrt{N(s)}}{1+N(s,a)}$$